

Sara Mueller

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 LinkedIn

EDUCATION

- **Doctor of Philosophy, Mathematics** *Expected Spring 2027*
University of Nebraska – Lincoln
Advised by Tom Marley
- **Masters of Science, Mathematics** *May 2023*
University of Nebraska – Lincoln
- **Bachelor of Arts in Mathematics, Cum Laude** *May 2021*
Wellesley College

RELEVANT COURSEWORK

- **Commutative Algebra I, Homological Algebra, Commutative Algebra II, Algebraic Geometry, Toric Geometry**
- *Reading Courses in:* **Galois Theory and Local Cohomology**

POSITIONS HELD

- **Graduate Teaching Assistant** *Aug. 2021 -*
University of Nebraska – Lincoln
- **Graduate Research Assistant** *Summers 2023, 2024*
University of Nebraska – Lincoln
 - Worked in Github, Webwork, and VSCode to make updates to online resources and course packets for first year mathematics courses
 - Designed course materials for undergraduate preparatory course for incoming engineering students.

TEACHING EXPERIENCE

- **Associate Convener**

Math 100A: Intermediate Algebra, *University of Nebraska – Lincoln*
Fall 2024 - Spring 2025

Math Resource Center and Learning Assistant, *University of Nebraska – Lincoln*
Fall 2023 - Spring 2024

Worked with MRC and LA directors, was responsible for designing schedule, and served as first point of contact for MRC and LA issue with graduate and undergraduate students

- **Instructor of Record**

Math 100A: Intermediate Algebra, *University of Nebraska – Lincoln*
Spring 2022, Fall 2024, Spring 2025

Math 101: College Algebra, *University of Nebraska – Lincoln*
Fall 2021

Math 101P: College Algebra Co-requisite, *University of Nebraska – Lincoln*
Fall 2023, Fall 2025, Fall 2026

Math 309: Introduction to Proof Writing, *University of Nebraska – Lincoln*
Summer 2025

- **Recitation/Lab Instructor**

Math 106: Calculus I, *University of Nebraska – Lincoln*
Fall 2020

Math 107: Calculus II, *University of Nebraska – Lincoln*
Spring 2021, Summer 2021

Math 221: Differential Equations, *University of Nebraska – Lincoln*
Spring 2024

Math 435: Math in the City, *University of Nebraska – Lincoln*
Spring 2024

UNDERGRADUATE MENTOR EXPERIENCE

- **First Year First Generation REU Mentor**

Summer 2026

University of Nebraska – Lincoln

- Mentored four incoming freshmen for a four week full-time internship, where students worked on a graph theory problem. Students finished the program with confidence talking about abstract math topics and presented their findings in a 30 minute talk, as well as in a poster format, both utilizing Beamer.

- **Directed Reading Program**

Fall 2025

University of Nebraska – Lincoln

- Mentored advanced mathematics major during weekly meeting, discussing an undergraduate Topology text and supplying and supporting problems to be worked on before and during the meetings.

PROFESSIONAL DEVELOPMENT

- **Pedagogy Course**

August 2022 – May 2023

Completed a year-long pedagogy course titled "Teaching and Learning Mathematics at the Post-Secondary Level". Topics included constructivism, communities of practice, humanistic teaching perspectives, and equitable teaching.

- **Preparing Future Faculty Fellowship**

May 2025 – December 2025

SKILLS

Programming Experience in: Macaulay2, MatLab, C++, Python

Typesetting Languages: $\text{\LaTeX}$, PG (Perl-based)

AWARDS

- **Don Miller Outstanding GTA Award**

Fall 2023 – Spring 2024

RESEARCH EXPERIENCE

- **Graduate Research**

Fall 2024 –

Currently studying Gorenstein homological dimension in the derived category

- **Undergraduate Research**

Fall 2020 – Spring 2021

Thesis Project in Schur Polynomials in 6 Vertex Lattice Models

Analyzing patterns in ice-type models with fewer paths than variables to determine a formula for representing these models as a combination of Schur polynomials

LEADERSHIP, SERVICE, AND VOLUNTEER ROLES

- **NCUWM Organizing Committee, Member**

2024-2026

The Nebraska Conference for Undergraduate Wisdom (prev. Women) in Mathematics is an annual mentorship and career event aimed at undergraduate women in mathematics. Panels, presentations, and networking opportunities are set up to encourage women towards mathematics-oriented graduate programs and career paths. My role as a member of the organizing committee involves participation in weekly planning meetings, coordinating volunteer participation, and managing the conference's social media accounts.

- **NCUWM, Volunteer**

2023-2025

As a volunteer at the Nebraska Conference for Undergraduate Women in Mathematics, I have served in various capacities during the conference, assisting as needed with set-up, clean-up, and organization, as well as advising and mentoring conference attendees.

- **UNL Math Day, Volunteer**

2022 - 2026

Math Day is an annual event designed to increase Nebraska high school students' enthusiastic participation in mathematics. As a volunteer, I helped facilitate competitions and program events.

- **Association for Women in Mathematics, Graduate Student Mentor**

2024-2026

I act as Graduate Student Mentor to University of Nebraska Lincoln's AWM Chapter, attending weekly meetings, assisting with planning and organizing events, and helping to foster an inclusive community for women mathematics students on campus.

- **First Year Graduate Student Mentor**

2023-2025

I act as a mentor to incoming first-year graduate students, meeting regularly with one new student each year to help ease their transition into the program.

- **UNL Math Circle**

2023-2024

Math Circle is a community outreach program coordinated with regional schools designed to bring fun and engaging math-related programming and activities to local elementary, middle, and high-school students. As a volunteer, I helped to design and lead after-school activities for several different age groups of students.

- **Girls Inc., Instructor**

2024

Girls Inc. is an organization which seeks to support girls and young women by building confidence, strength, and knowledge through girl-focused summer camp programming. As an instructor, I planned and taught lessons and activities focused around fun, real-world applications of mathematics, including tessellations and encryption, culminating in a math-focused escape room challenge.

TALKS

- **Great Plains Alliance** Wayne State College, Wayne, NE

March 2024

– Gave talk on p -adics to undergraduate math students

- **Commutative Algebra Research Seminar** University of Nebraska-Lincoln

February 2025

– Titled “Totally Reflexive Modules and G-dimension”, this talk gave an introduction to this homological dimension, through examples and proofs, as well as some open problems associated to the subject

- **Commutative Algebra Research Seminar** University of Nebraska-Lincoln

November 2025

– Titled “Homological Dimensions, Regularity, and New Intersection Theorem”, this talk reviewed two main problems in my current research.